



Original Article

Validation and development of population pharmacokinetic model of vancomycin using a real-world database from a nationwide free web application

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ABSTRACT

Introduction: Vancomycin requires a population pharmacokinetic (popPK) model to estimate the area under the concentration-time curve (AUC), and an AUC-guided dosing strategy is necessary. This study aimed to develop a popPK model for vancomycin using a real-world database pooled from a nationwide web application (PAT).

Methods: In this retrospective study, the PAT database between December 14, 2022 and April 6, 2023 was used to develop a popPK model. The model was validated and compared with six existing models based on the predictive performance of datasets from another PAT database and the Kumamoto University Hospital. The developed model determined the dosing strategy for achieving the target AUC.

Results: The modeling populations consisted of 7146 (13,372 concentrations from the PAT database), 3805 (7540 concentrations from the PAT database), and 783 (1775 concentrations from Kumamoto University Hospital) individuals. A two-compartment popPK model was developed that incorporated creatinine clearance as a covariate for clearance and body weight for central and peripheral volumes of distribution. The validation demonstrated that the popPK model exhibited the smallest mean absolute prediction error of 5.07, outperforming others (ranging from 5.10 to 5.83). The dosing strategies suggested a first dose of 30 mg/kg and maintenance doses adjusted for kidney function and age.

Conclusions: This study demonstrated the updating of PAT through the validation and development of a popPK model using a vast amount of data collected from anonymous PAT users.

1. Introduction

Vancomycin, a glycopeptide antibiotic, was initially used to treat infections caused by gram-positive bacteria, specifically methicillin-resistant *Staphylococcus aureus* (MRSA) [1]. Guidelines recommend

therapeutic drug monitoring (TDM) due to increasing evidence of dose-dependent nephrotoxicity and clinical efficacy [2–4]. The recommended therapeutic range for the area under the concentration-time curve (AUC) is between 400 and 600 $\mu\text{g}\cdot\text{h}/\text{mL}$, assuming a minimum inhibitory concentration (MIC) of 1 $\mu\text{g}/\text{mL}$ [3–5]. To estimate the AUC, a software utilizing Bayesian forecasting and a population

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Abbreviations

AUC	area under the concentration-time curve
AWS	Amazon Web Services
CCr	creatinine clearance
CL	clearance
FOCE-I	first order conditional estimation with interaction
IIV	Interindividual variability
MIPD	model-informed precision dosing
MRSA	methicillin-resistant <i>Staphylococcus aureus</i>
PAT	practical area under the concentration-time-guided therapeutic drug monitoring
popPK	population pharmacokinetic
Q	intercompartmental clearance
RV	residual variability
TDM	therapeutic drug monitoring
V1	central compartmental volume of distribution
V2	peripheral compartmental volume of distribution

pharmacokinetic (popPK) model was introduced [6–11]. An individual can apply a well-fitted popPK model as *a priori* probability, and models for various races have been reported [11–16]. However, existing popPK models have been developed and evaluated based on clinical studies with a relatively small number of participants, with 85 % of the models based on fewer than 200 participants [11]. Thus, there is room for further improvement of popPK models.

To promote AUC-guided dosing for vancomycin, a web application called PAT (named after the initial abbreviations of Practical AUC-guided TDM) has been developed [8]. The input data were collected and pooled to monitor appropriate usage and identify areas for improvement. Notably, PAT can be used to create a real-world database involving vancomycin dosing schedules and measured concentrations, including records of time, with less effort. Therefore, we hypothesized that a new popPK model could be developed in a novel manner using a real-world database. If proven valid, this approach could have great potential not only for vancomycin, but also for other drugs and cases. Therefore, the objective of this study was to develop an original model that leverages a real-world clinical practice database pooled using PAT.

2. Materials and methods

2.1. Study Design, data source, and ethics

This was a retrospective study. The real-world clinical practice database was pooled to monitor appropriate usage and identify areas for improvement, specifically for clinical purposes and not for clinical research. A privacy policy based on the Japanese regulations was introduced. The data were stored in CSV format on Amazon Web Services (AWS) through PAT usage, and access to the data was controlled by the first author. As the data lacked an identifier for specifying an institute, area, or individual, every record was unverifiable. The data source for the development of the popPK model was retrieved from a real-world clinical practice database from December 14, 2022 to April 6, 2023. Two data sources were used to evaluate the popPK model. One was retrieved from a real-world clinical practice database from April 7 to May 6, 2023, and the other was created through a retrospective investigation of TDM data at Kumamoto University Hospital between January 1, 2017 and December 31, 2021. Individual dosing regimens, history of measurements, and laboratory test results were thoroughly reviewed from electronic medical records. Concentrations were measured using the Cobas c 501 (Roche Diagnostics) analyzer, with a coefficient of variation of less than 15 % and a calibration range of 2.5–100 µg/mL. Values below the calibration range were excluded from the analysis.

The Institutional Review Board (IRB) of Kumamoto University Hospital approved the use of PAT data and clinical TDM data for this study (approval numbers 2542 and 2,413, respectively). The IRB waived the requirement for signed informed consent from individuals; instead, we offered an option on our website for individuals to opt out of participating in the study. This study was conducted according to the Declaration of Helsinki.

2.2. PAT aspects

PAT, a web application for the AUC-guided dosing of vancomycin, was developed using the R programming language (<https://www.r-project.org/>) and the Shiny package (<https://cran.r-project.org/web/packages/shiny/index.html>) as a web application framework. It was deployed on a cloud server hosted on the AWS and made available for use in October 2020 (sample website: <https://pharmacokinetic-simulation.shinyapps.io/app-ver1/>). An adult popPK model by Yasuhara et al. was registered [12]. Each user is assigned a specific identifier number at the time of access.

Input data is recorded at the time of performing Bayesian forecasting [17] and includes the following items: the specific identifier, a time-stamp for Bayesian forecasting, the purpose of usage, the popPK model, the MIC value, sex, age, body weight, serum creatinine level, creatinine clearance value calculated by the Cockcroft-Gault equation [18] or measured by pooled urine (CCr), checkboxes representing excessive muscle loss categories (myogenic disorder [muscular dystrophy], neurogenic disorder [spinal cord injury with complete paralysis, cerebral palsy, amyotrophic lateral sclerosis], inflammatory disorder with prolonged bed rest [polymyositis, dermatomyositis]), checkboxes representing other muscle loss factors (prolonged bed rest prior to hospitalization, stay longer than 14 days in an intensive care unit) [19–22], serum albumin level, body height, serum cystatin C level, presence in ICU or pediatric ICU, burn, insipidus, increase or decrease in serum creatinine levels, risk factors for nephrotoxicity, dosing regimen, and measured concentration over time after dose.

2.3. Data curation

A crude pooled real-world clinical practice database was curated using the following rules: Duplicate records with the same specific identifier number were removed, and the record with the latest time-stamp was retained. Records lacking information on sex, age, body weight, serum creatinine level, dosing regimen (dose, infusion time, and number of doses), and measurements were eliminated. Additionally, records that could not be used for Bayesian forecasting (e.g., the recorded time of the first dose after the first measurement) were excluded.

Records that showed identical conditions before and after 30 days, based on specific combinations, were excluded. These combinations included 1) age, sex, body weight, and serum creatinine level; 2) sex, age, first dose, and first concentration; 3) sex, age, body weight, and first concentration; 4) date, age, and body weight; 5) first and subsequent doses, and first concentration; 6) date, first, and second concentrations; and 7) unchanged first values for a body weight of 60 kg, age of 60 years, and serum creatinine level of 0.8. Additionally, records that seemed excessive, such as serum creatinine levels >10 mg/dL, body weight >200 kg, creatinine clearance (CCr) of zero, a dose <2.5 mg/dose or >5000 mg/dose, or >40 mg/kg/dose, and a concentration of 0 or >100 µg/mL, were excluded. If the recorded CCr value differed from the calculated result, the serum creatinine level was inversely calculated using the Cockcroft equation to represent the measured value.

2.4. Population pharmacokinetic modeling and validation

The modeling process is documented in [Supplementary Text S1](#). The developed popPK model was validated using *a priori* probability with the

Table 1
Demographics of prepared dataset.

Character	Modeling dataset (n = 7167)			Validation dataset 1 from PAT usage (n = 3805)			Validation dataset 2 from Kumamoto Univ. Hosp. (n = 783)		
	No. of observation	Mean (SD)	Median [Min, IQR, Max]	No. of observation	Mean (SD)	Median [Min, IQR, Max]	No. of observation	Mean (SD)	Median [Min, IQR, Max]
Age, years	7167 (100 %)	70.1 (16.5)	74 [15, 60–82, 106]	3805 (100 %)	71.3 (16.3)	75 [15, 62–83, 105]	783 (100 %)	64.1 (15.9)	67 [18, 57–75, 95]
Body weight, kg	7167 (100 %)	55.7 (14.8)	54 [16.3, 45–64, 151.3]	3805 (100 %)	54.9 (14.9)	53.2 [15, 45.0–63.0, 182.0]	783 (100 %)	57.6 (13.9)	56.5 [24.0, 48.2–65.0, 159.7]
eGFR, mL/min/1.73m2	7167 (100 %)	77.2 (43.4)	71.2 [3.4, 50.5–95.8, 808.2]	3805 (100 %)	76.7 (40.3)	71.2 [4.9, 50.7–95.3, 371.0]	783 (100 %)	91.7 (61.8)	82.3 [6.5, 60.5–107.5, 845.5]
CCr, mL/min	7167 (100 %)	67.5 (40.1)	58.5 [2.0, 39.8–85.2, 389.4]	3805 (100 %)	65.5 (38.9)	57.6 [1.3, 38.5–82.6, 471.9]	783 (100 %)	92.3 (56.0)	82.2 [7.7, 56.9–115.1, 644.8]
First dose, mg/kg	7167 (100 %)	21.4 (7.0)	22.2 [1.6, 15.6–26.8, 40.0]	3805 (100 %)	22.1 (6.9)	23.3 [1.4, 16.9–27.3, 39.9]	783 (100 %)	20.9 (6.2)	20.6 [6.5, 16.1–25.8, 43.0]
Maintenance dose ^a , mg/kg/dose	7167 (100 %)	14.9 (5.5)	14.3 [1.6, 10.8–18.2, 39.4]	3805 (100 %)	14.8 (5.6)	14.2 [1.4, 10.7–18.1, 39.7]	783 (100 %)	15.0 (4.5)	14.7 [3.8, 11.8–17.9, 39.5]
No. of concentration <2.5 h after dose	13,372			7540			1775		
2.5–< 3.5 h after dose	1442 (10.8 %)			899 (11.9 %)			133 (9.7 %)		
Measured concentration, µg/mL	1098 (8.2 %)			758 (10.1 %)			40 (2.3 %)		
Measured time after dose, h	7167 (100 %)	17.4 (8.9)	16 [0.1, 11.3–21.7, 72.8]	3805 (100 %)	17.5 (8.8)	15.8 [0.1, 11.3–21.8, 92.8]	783 (100 %)	15.3 (6.6)	14.4 [2.2, 10.7–18.7, 70.2]
Body height, cm	7167 (100 %)	12.8 (8.3)	12.0 [0.5, 8.5–12.0, 49.5]	3805 (100 %)	12.3 (8.1)	12.0 [1.0, 8.0–12.0, 49]	783 (100 %)	11.8 (9.5)	11.0 [0.5, 8.0–12.0, 191.9]
Serum albumin, g/dL	2532 (35.3 %)	160.0 (10.8)	160.0 [57.9, 153.0–168.0, 189.0]	1369 (36 %)	160.2 (11.4)	160.0 [1.6, 153.0–160.2, 192.0]	783 (100 %)	160.0 (9.2)	160.0 [128.0, 153.5–166.0, 188.0]
Cystatin C, µg/mL	2535 (35.4 %)	2.4 (0.6)	2.4 [0.1, 2.0–2.8, 5.1]	1349 (35.5 %)	2.4 (0.6)	2.3 [0.5, 2–2.8, 5.4]	783 (100 %)	2.7 (0.7)	2.7 [0.6, 2.2–3.1, 4.8]
Any muscle loss, n (%)	116 (1.6 %)	1.41 (0.54)	1.34 [0.46, 1.04–1.61, 3.43]	48 (1.3 %)	1.52 (0.97)	1.34 [0.67, 0.98–1.59, 7.00]	0 (0 %)	–	–
Excessive, n (%)	811 (11.3 %)			423 (11.1 %)			224 (28.6 %)		
Others, n (%)	584 (8.1 %)			286 (7.5 %)			137 (17.5 %)		
Intensive care unit, n (%)	284 (4 %)			162 (4.3 %)			91 (11.6 %)		
Burn, n (%)	318 (4.4 %)			156 (4.1 %)			110 (14 %)		
Inspidus, n (%)	13 (0.2 %)			13 (0.3 %)			0 (0 %)		
	0 (0 %)			0 (0 %)			0 (0 %)		

eGFR, estimated glomerular filtration rate; CCr, creatinine clearance; IQR, interquartile range.

^a Maintenance dose represents any dose after first dose.

datasets provided in the subsequent phase of PAT usage (validation dataset 1) and Kumamoto University Hospital (validation dataset 2) mentioned above (please see Study Design, Data Source, and Ethics in Methods). Mean prediction error (µg/mL) and mean absolute prediction error (µg/mL) were used to evaluate bias and accuracy, respectively, using the following equations:

$$\text{Mean prediction error} = 1 \bigg/ n \sum_{i=1}^n (\text{prediction}_i - \text{observation}_i)$$

$$\text{Mean absolute prediction error} = 1 \bigg/ n \sum_{i=1}^n |\text{prediction}_i - \text{observation}_i|$$

(where n represents the total number of individuals in the dataset). The predictive performance of the developed final popPK model was compared with existing popPK models based on the following criteria: 1) models employed in the PAT software [12], 2) models developed by a prospective study based on rich sampling, 3) models developed by a retrospective study with sparse sampling but employing more than 100 participants, and 4) models developed during the search period between 2010 and 2020. Models were excluded if they incorporated covariates other than sex, age, body weight, serum creatinine levels, or CCr.

2.5. Suggested dosing

First and maintenance doses were calculated using the demographics of the dataset, which included both the modeling and validation groups, using their respective simulated dosing schedules. Pharmacokinetic parameters for individuals were estimated through random resampling, based on the distribution of the final popPK model. Concentration-time curves were simulated in R using the desolve function to solve the differential equations, with a calculation interval of 0.1 h. The dosing schedule consisted of a first (loading) dose of 25, 30, or 35 mg/kg, followed by a maintenance dose to achieve an AUC of 500 µg·h/mL at a steady state. The maintenance dose was typically divided into two doses administered 12 h (h) apart, starting 12 h after the first dose. However, if the maintenance dose was less than 17.5 mg/kg/day, it was administered once daily, 24 h after the first dose. Since the AUC on day 2 was suggested to be in the range of 389.8–515.7 µg·h/mL [23], the suggested first dose was based on the probabilities of achieving AUC values in the range of 400–500 µg·h/mL on day 1 or 2. The AUC on day 2 was calculated by multiplying the mean simulated concentrations (240 points over 24 h, sampled every 0.1 h) by 24. The maintenance doses were investigated at medians of CCr (mL/min) (corresponding to the different ranges) of: >168, 159 (150–168), 135 (120–149), 105 (90–119), 75 (60–89), 52.5 (45–59), 37.5 (30–44), 22.5 (15–29), and 10 (5–14). The maintenance doses at steady state were calculated by

Table 2
Final population pharmacokinetic model.

Fixed effect parameter	Value	%RSE (η or ε shrinkage)	Estimated parameter by Bootstrap method using 1000 resamples (505 successful)	
			Median	95 % Confidence Interval
IF CCr >01 THEN FAC _{CCr} = 01 ELSE FAC _{CCr} = CCr CL (L/h) = 02 × (FAC _{CCr} /120) ⁰³				
01	167.8	1.1 (–)	167.8	142.3–193.3
02	4.9	0.8 (–)	4.9	4.8–5.0
03	0.959	0.99 (–)	0.960	0.938–0.983
V1 (L) = 04 × (body weight/60) ⁰⁵				
04	32.1	3.2 (–)	32.4	29.7–34.9
05	0.532	12.2 (–)	0.529	0.417–0.638
Q (L/h) = 06rowhead				
06	8.33	4.4 (–)	8.24	7.23–9.25
V2 (L) = 07 × (body weight/60) ⁰⁸				
07	58.2	3.2 (–)	58.3	54.9–61.2
08	0.348	17.6 (–)	0.360	0.210–0.511
Random effect parameter				
ω_{CL}^2	0.136	3.2 (16.7)	0.135	0.126–0.144
ω_{V1}^2	0.111	13.8 (73.8)	0.108	0.077–0.163
ω_{V2}^2	0.530	5.2 (44.8)	0.536	0.473–0.595
σ^2	0.0329	4.1 (31.4)	0.0327	0.0302–0.0355

Interindividual and residual variabilities were in exponential distribution.
The unit for body weight was kg.
CCr, creatinine clearance (mL/min).

multiplying the simulated clearances by 500. Conventional maintenance doses were determined by rounding these doses to practical values. The probability of target attainment for the AUC at steady state, calculated by the ratio of dose per day to clearance, was estimated using Monte Carlo simulations in R software, based on 1000 resamplings. The *rnorm* and *runif* functions in R were utilized to generate normally distributed random numbers for the mean and standard deviation (SD) of the clearance, and uniform random numbers for CCr within the specified range. Additionally, the appropriateness of using estimated glomerular filtration rate (eGFR) (mL/min/1.73 m²)-guided maintenance doses was investigated through Bayesian posterior estimates of clearance for individuals, derived from NONMEM analysis.

3. Results

3.1. Data curation

The three datasets are listed in Table 1. The histograms of these demographics are shown in Supplementary Figs. S1–3. The crude modeling dataset consisted of 103,970 records, which were curated by removing them in accordance with predefined exclusion rules. The curation process resulted in 7167 (96.3 %) records, with 13,372

confirmed concentrations for adults (≥15-year-old) in the modeling dataset. The crude validation dataset from the PAT consisted of 45,652 records, which were curated by removing those in accordance with the predefined exclusion rules. The curation process finally confirmed 3805 (97.7 %) records with 7540 concentrations for adults. The validation dataset from the clinical TDM data at Kumamoto University Hospital confirmed 783 records with 1775 adult concentrations.

3.2. Population pharmacokinetic modeling and validation

The results of the modeling process are documented in [Supplementary Text S2](#). The final model was presented in Table 2, including the median and corresponding 95 % CI range estimated by the bootstrap method using 1000 resamples. The goodness-of-fit diagnostic and prediction-corrected visual predictive check plots are shown in Fig. 2.

We selected six existing popPK models to validate the final model using *a priori* probability [12,24–28], as presented in [Supplementary Table S7](#). The prediction and absolute prediction errors in the validation datasets are plotted in Fig. 3 as representative mean (SD) values. In the results for the total datasets, the developed final popPK model (ORIGINAL) showed the second smallest prediction errors in the validation datasets 1 and 2 (Fig. 3a), while showing the smallest absolute prediction errors in the validation datasets 1 and 2 (5.07, vs. others in the range of 5.10–5.83, Fig. 3b). These findings were consistent when the validation was stratified by CCr >85 mL/min, which was the cut-off value in the popPK model of Yasuhara et al. (Fig. 3c and d) [12]. However, when the validation was stratified by CCr <85 mL/min, ORIGINAL showed the smallest prediction error in validation dataset 1 but ranked 4th in validation dataset 2 (Fig. 3e). The absolute prediction error was similar to that of the other models (Fig. 3f).

3.3. Dosing strategy

An investigation of the first dose is shown in Fig. 4. Increasing the first dose from 30 to 35 mg/kg resulted in a slight increase in the probability of achieving an AUC in the range of 400–500 µg·h/mL on day 2 (from 26.1 % to 29.1 %), indicating a mere 3.0 % increase. However, the probability of achieving an AUC above 500 µg·h/mL increased from 20.4 % to 34.3 %.

Maintenance doses stratified by CCr are shown Table 3. Because of the upper limit for CCr of 168 mL/min, the maximum dose was estimated to be 3614 mg/day. When in CCr <120 mL/min, 2000 mg/day was conventionally suggested due to the estimated value of 2156 mg/day in CCr range of 90–119 mL/min. Investigation for appropriateness of eGFR-guided maintenance dose was resulted in Fig. 5 (the ratio of CL/eGFRcre or CL/CCr, along with age and body weight, respectively) and [Supplementary Fig. S9](#) (a dose–eGFRcre plot). As illustrated in Fig. 5a,

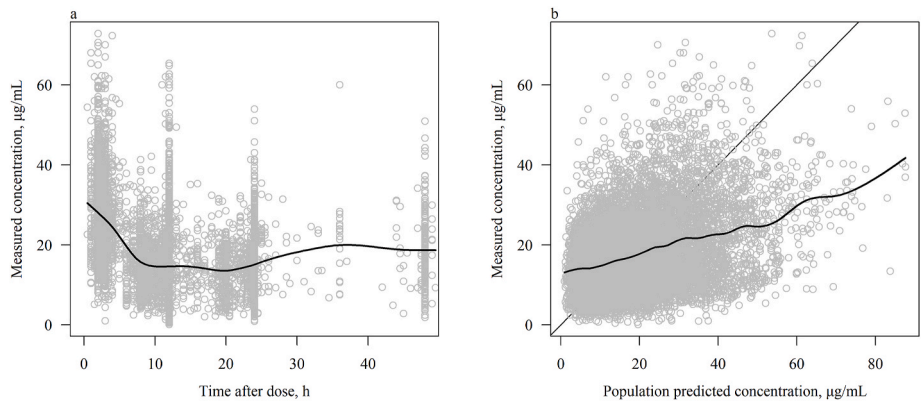


Fig. 1. Measured Concentrations. a. Measured concentration over time after dose b. Measured-predicted concentration. The prediction was derived from the base model developed in this study. Bold lines represent the smoothed line.

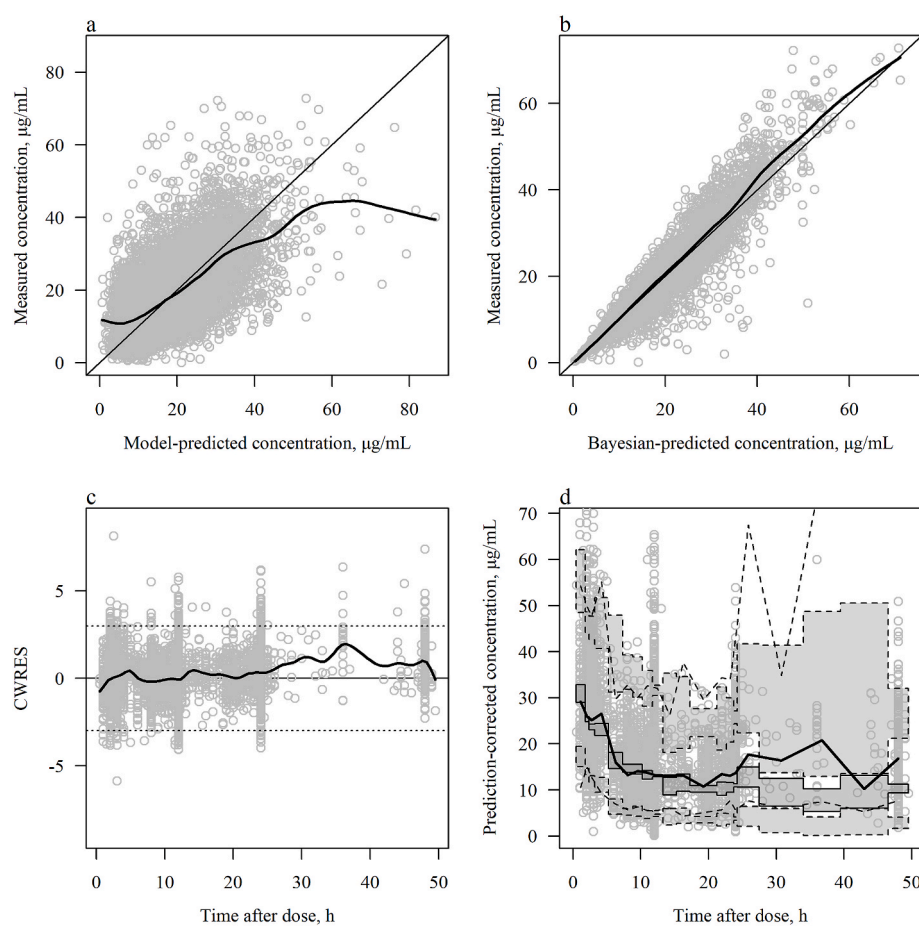


Fig. 2. Internal evaluation of developed final population pharmacokinetic (popPK) model. a. Model-predicted concentration (calculated as PRED by NONMEM) was derived from the developed final popPK model. The plot on logarithmic axes is shown in [Supplementary Fig. S7](#) b. Bayesian-predicted concentration (calculated as IPRED by NONMEM) was derived from NONMEM calculation. The plot on logarithmic axes is shown in [Supplementary Fig. S7](#) c. CWRES represents conditional weighted residuals. The dotted lines represent ± 3 . d. Prediction-corrected visual predictive check. The bold line represents the observed median, dashed lines represent the 95th percentile range, and grey areas represent the 95 % confidence intervals of the corresponding lines.

the CL/eGFRcre ratio decreased with age and altered with body weight. In contrast, the CL/CCr ratios in aging and body weight were flat, as shown in [Fig. 5c](#) and [d](#), respectively. Maintenance doses stratified by eGFR, and considering predictive variations based on increasing age, are shown in [Supplementary Table S8](#).

4. Discussion

We developed a popPK model for vancomycin using an unverifiable real-world clinical practice database obtained from a nationwide web application. Although the large dataset might contain some recording errors, as suggested by the low success rate of bootstrap estimation (50.5 %, [Table 2](#)), the parameters estimated in our study were similar to those reported in previous studies ([Supplementary Table S7](#)). Frequent sampling during the peak period (10–20 % in all observations, as shown in [Table 1](#) and [Fig. 1a](#)) allowed us to estimate the volume of distribution using IIV. However, despite the large population size of 7167 individuals, our study could not estimate the IIV for Q, possibly because of the insufficient number of observations for accurate estimation, which is consistent with previous studies ([Supplementary Table S7](#)) [11–16, 24–28]. The large shrinkage for the volume of distribution, as described in [Table 2](#), might also indicate that the model is over-parameterized. Considering the study by Lim et al. even with a relatively small population size ($n = 20$), rich sampling allowed for accurate estimation [24]. Although an extremely large population size may enable an estimation, the accuracy of such estimates is questionable. Underprediction around

30 h after dosing may be a concern, as indicated by [Fig. 2a](#), [b](#), and [2d](#). This could be due to the relatively smaller number of observations. Because the bias observed 40 h post-dose becomes smaller. Although PAT software is generally not permitted for patients undergoing extracorporeal blood purification therapies like hemodialysis, this study may have inadvertently included such patients due to its design and the specified minimum CCr of 1.3 mL/min, as shown in [Table 1](#). Consequently, we developed a popPK model including only patients with a CCr < 15 mL/min ($n = 6,978$, 97.4 %). The parameters for the fixed and random effects in the popPK model closely matched the original values (θ_1 : 167.8 vs. 168.0; θ_2 : 4.9 vs. 4.9; θ_3 : 0.959 vs. 0.955; θ_4 : 32.1 vs. 32.4; θ_5 : 0.532 vs. 0.534; θ_6 : 8.33 vs. 8.26; θ_7 : 58.2 vs. 59.8; θ_8 : 0.348 vs. 0.329; ω_{CL}^2 : 0.136 vs. 0.135; ω_{V1}^2 : 0.111 vs. 0.108; ω_{V2}^2 : 0.530 vs. 0.557; σ^2 : 0.0329 vs. 0.0328). Considering the inclusion of the small number of patients with a CCr < 15 mL/min, their potential impact on the study was considered minimal.

The bias explained by the mean prediction error was not significant when validating all observations ([Fig. 3a](#)), except for the model by Medellín-Garibay et al. [26]. Their model equation for estimating V2 was $5.86 \times \text{body weight}$, which is considerably larger than other reported values ([Supplementary Table S7](#)), resulting in significant bias. We speculate that the value of Q (0.81 L/h) and the coefficient (5.86) may have been placed inversely. In a validation limited to CCr > 85 mL/min, the model reported by Yasuhara et al. showed significant bias, which could be attributed to fixing CL at 3.51 L/h within the range of CCr [12]. The other four models and the validation limited to CCr < 85

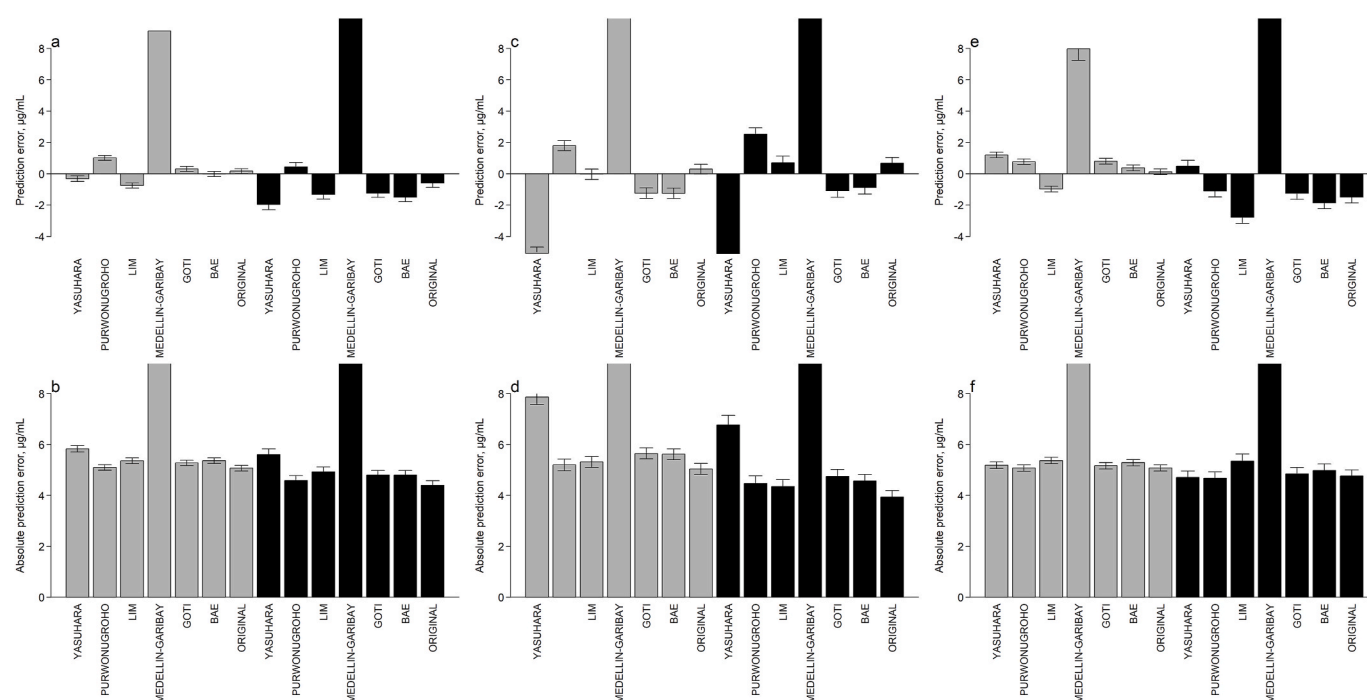


Fig. 3. Validation of final population pharmacokinetic (popPK) model. a, c, and e. The height of bars represents mean prediction errors (for the dataset of total, CCr >85 mL/min, CCr <85 mL/min, respectively). b, d, and f. The height of bars represents mean absolute prediction errors (for the dataset of total, CCr >85 mL/min, CCr <85 mL/min, respectively). Grey and black bars represent results from the validation dataset 1 (from PAT usage) and 2 (from Kumamoto University Hospital), respectively. The error bars exhibit their corresponding 95 % confidence intervals.

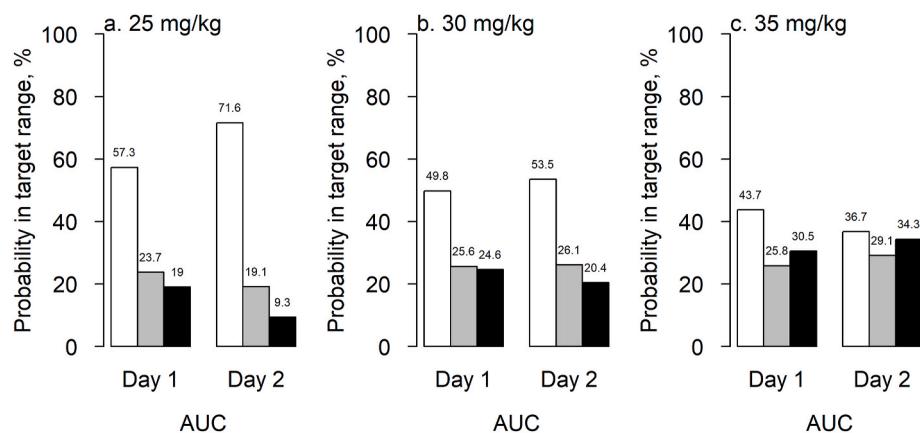


Fig. 4. Probability in target range of area under the concentration-time curve (AUC) by different first doses. White, grey, and black bars represent AUC of <400, 400–500, and >500 µg·h/mL, respectively. The probability of achieving an AUC in the range of 400–500 µg·h/mL on day 1 ranged from 23.7 % to 25.8 %. However, a larger range of probabilities was observed for AUC values below 400 µg·h/mL (43.7 %–57.3 %) and above 500 µg·h/mL (19.0 %–30.5 %). On day 2, the probability of achieving an AUC in the range of 400–500 µg·h/mL ranged from 19.1 % to 29.1 %, while the probabilities for AUC values below 400 µg·h/mL and above 500 µg·h/mL were 36.7 %–71.6 % and 9.3 %–34.3 %, respectively.

mL/min presented comparable biases. Regarding the accuracy explained by the mean absolute prediction error, the developed final model demonstrated the smallest mean absolute prediction error in the validation, limited to CCr >85 mL/min and all observations. However, patients with kidney dysfunction should receive a qualified dosing strategy based on various popPK models (Fig. 3f). In summary, the validation demonstrated that there was no reason to not use the developed final model as *a priori*.

We investigated the first dose targeting an AUC range of 400–500 µg·h/mL on days 1 and 2 [23]. The difference in achieving the target AUC on day 1 was small among the first doses (Fig. 4). This might be reasoned by the target AUC at steady state of 500 µg·h/mL. However, the probability of achieving an AUC >500 µg·h/mL on day 1 remarkably

increased. These results suggest that the first-dose decision could be based on the AUC on day 2. The probability of achieving the target AUC range increased from 19.1 % (25 mg/kg) to 26.1 % (+7.0 %, 30 mg/kg), with a smaller increase to 29.1 % (+3.0 %, 35 mg/kg). In contrast, the probability of an AUC >500 µg·h/mL on day 2 significantly increased from 9.3 % (25 mg/kg) to 20.4 % (30 mg/kg) and further to 34.3 % (35 mg/kg). Considering the small increase in the probability of achieving the target AUC range compared with the large increase in the probability of a toxic AUC, the present study suggests a first dose of 30 mg/kg. This dose is recommended by the guidelines [2–4]. Although guidelines in the USA have introduced 35 mg/kg as the first dose [3], which may help avoid subtherapeutic exposure, further studies are warranted to evaluate its safety.

Table 3
Suggested maintenance dose for area under the concentration-time curve (AUC).

Creatinine clearance, mL/min	Maintenance dose for AUC of 500 µg·h/mL	Conventional suggested maintenance dose	
		Dose	Probability of AUC within 400–600 µg·h/mL
>168	3614	1000–1250 mg/dose, q 8 h	41.0 %
150–168	3209		
120–149	2743		
90–119	2156	1000 mg/dose, q 12 h	41.0 %
60–89	1561	750 mg/dose, q 12 h	40.0 %
45–59	1109	1000 mg/dose, q 24 h	42.1 %
30–44	803	750 mg/dose, q 24 h	43.1 %
15–29	492	500 mg/dose, q 24 h	36.2 %
5–14	175	150–200 mg/dose, q 24 h	32.2 %

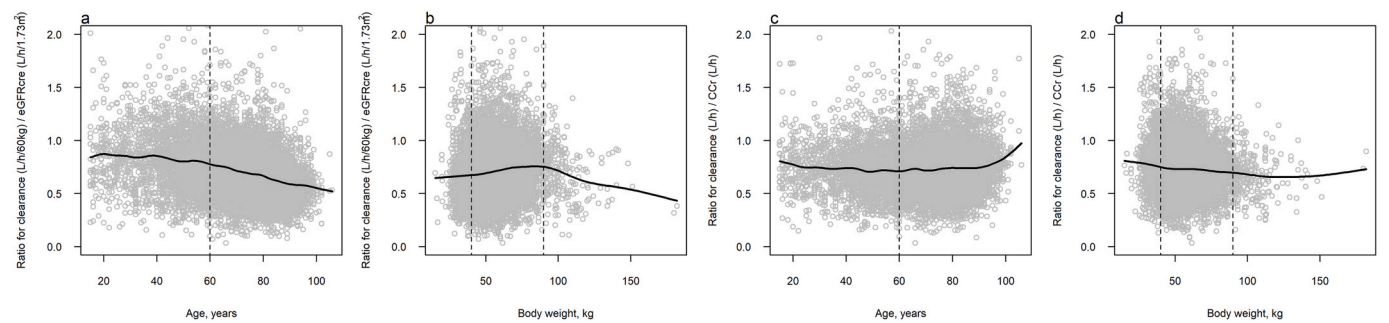


Fig. 5. Ratio for clearance against estimated glomerular filtration rate using serum creatinine level (eGFRcre) or creatinine clearance (CCr) along with age and body weight. Dash lines for panel a and c represent 60 years old, panel b and d represent 40 or 90 kg of body weight, respectively. Bold lines represent their corresponding smooth lines.

In the investigation of the maintenance dose for an AUC of 500 µg·h/mL, dosing according to CCr was suggested as a strategy is widely practiced (Fig. 5, Table 3). Particularly, the population in this study demonstrated a decreasing CL/eGFR ratio with increase in age (Fig. 5a). This finding suggests a potential overestimation of eGFR after CL in the elderly population. Interestingly, the CL/CCr ratio remained constant with age and body weight (Fig. 5c and d). Considering that the allometric scaling factor of CCr for CL was 0.959 and the component of age in the Cockcroft-Gault equation was (140 - Age), the impact of age on vancomycin CL was greater for CCr than that for eGFR. Therefore, stratifying the investigation of the maintenance dose by age is reasonable but the threshold age could not be estimated in this study. Because the final popPK model appears to fit well in elderly patients, as illustrated in Supplementary Fig. S1, the final model-informed dosing may be challenging for relatively younger individuals. For these relatively younger individuals, a dose of 20 mg/kg every 8 h can be suggested. This dosage, which is considered in the guidelines, is higher than the recommended 20 mg/kg every 8 h for patients with normal kidney function [2–4].

This study has several limitations. First, the study utilized an unverifiable real-world database, which potentially led to misleading model development owing to possible intentional errors in the data input. As demonstrated in this study, validation based on clinical studies is warranted. Second, this methodology cannot identify novel covariates in the popPK model, which require rigorous clinical research. Third, this approach involves non-uniform measurement methods for laboratory tests and drug concentrations. Consequently, this may result in the derivation of a portion of the components of real-world residual variability.

Nevertheless, this methodology holds great potential for future modeling and refinement within the framework of model-informed precision dosing (MIPD) [29–31]. With the advancement of technology (e.g., artificial intelligence), the most efficient approach would be to complete the upcycle using a real-world database without relying solely on clinical research. Therefore, it is important to demonstrate the effectiveness of this method using other pharmacotherapies, models,

and populations.

In conclusions, we demonstrated that a vancomycin popPK model can be validated and developed using a vast amount of data collected from anonymous PAT users in clinical practice. This methodology has the potential to contribute to the upcycling of MIPD in the upcoming era.

Authorship statement

KO was responsible for the organization and coordination of the trial. KO, KM, KS, AS, HK, YTakahashi, TKatanoda, YH, HJ, HS, YTakesue, and TKimura developed the trial design. KO drafted the article, and all authors revised it critically for important intellectual content. TKimura was the chief investigator and responsible for the data analysis. All authors approved the final manuscript. All authors meet the ICMJE authorship criteria.

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Declaration of competing interest

None.

Acknowledgements

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jiac.2024.05.014>.

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